

MTH6113 Assessment 2

Stock Returns, Prices and Lognormal Model Analysis

This report analyses three stocks chosen based on the first letters of each team member's first name: K Wave Media (chosen by Kenekchukwu), Meta (chosen by Morgan) and L'Oréal (chosen by Lisa). For each stock, 251 trading days of daily closing prices were collected. We compute daily log-returns, plot prices and returns, and simulate lognormal stock price paths using the empirical mean and standard deviation of each stock's log-returns. Finally, we evaluate the quality of the lognormal model and discuss its limitations.

Data Summary and Empirical Statistics

The table below summarises the key empirical statistics for each stock over the sample period (approximately one year of daily data):

Stock	Daily Mean Log-Return (μ)	Daily Std Dev (σ)	Ann. Mean Return	Ann. Volatility	Skewness
K Wave Media	-0.008903	0.1620	-2.2436	2.5710	-0.68
Meta	+0.000211	0.0242	+0.0531	0.3844	+0.66
L'Oréal	-0.000014	0.0256	-0.0034	0.4071	-0.39

The three stocks represent very different profiles. K Wave Media is a small-cap, highly volatile stock: its annualised volatility exceeds 250% and its price fell from a high of \$9.50 to \$0.69 over the year, generating a strongly negative average return. Morgan's stock is a large cap security with moderate volatility (38% annualised) and a mildly positive drift. L'Oréal is a large international company with similar volatility (41% annualised) and near-zero average daily return over the period, suggesting relative stability.

Returns and Prices: Observations

K Wave Media

The price chart for K Wave Media shows an extreme downward trend, with the stock losing roughly 93% of its value from its peak during the observation window. The daily log-returns exhibit very large swings, regularly exceeding $\pm 20\text{--}30\%$, which is atypical for most equity securities. This reflects the highly speculative nature of small-cap stocks. The return series shows clear signs of volatility clustering: periods of large moves tend to be followed by further large moves. The negative skewness (-0.68) indicate that large negative returns occur far more frequently than a normal distribution would predict.

Meta

This stock exhibits a more typical equity profile. Prices fluctuate within a range of approximately \$484 to \$790 over the year. The daily returns are relatively small, centred near zero, with occasional outliers. The positive skewness ($+0.66$) suggests a distribution

with a long right tail, meaning extreme positive moves occur more frequently than a normal model predicts, while most returns remain clustered at lower values

L'Oréal

L'Oréal's price action fluctuates within a stable range of €364 to €476, exhibiting moderate volatility without a defined trend. While daily returns are tightly clustered around zero, the negative skewness (-0.39) indicates a modest left-tail bias, suggesting that downside outliers occur with slightly more magnitude or frequency than equivalent moves to the upside.

Lognormal Model: Simulation

The lognormal model assumes that stock prices follow a Geometric Brownian Motion (GBM):

$$S(t+1) = S(t) \cdot \exp\left((\mu - \sigma^2/2) + \sigma \cdot Z\right)$$

where $Z \sim N(0,1)$ is a standard normal random variable, μ is the empirical daily mean log-return, and σ is the empirical daily standard deviation. The term $(\mu - \sigma^2/2)$ is the drift correction that ensures the expected price grows at rate μ rather than $\mu + \sigma^2/2$.

In the Excel workbook, we generated 251 simulated daily prices for each stock starting from the first observed price, using `NORM.S.INV(RAND())` to draw standard normal random variates. The simulation was repeated to illustrate the stochastic nature of the paths.

For K Wave Media, the high volatility ($\sigma \approx 0.162$ per day) dominates the simulation entirely: the simulated paths exhibit enormous variation, with some trajectories rising sharply and others collapsing to near zero within weeks. For the Morgan and L'Oréal stocks, simulated paths are more contained, broadly tracking the range of the observed price series, though individual paths can still diverge substantially from the actual price history.

Drawbacks of the Lognormal Model

Although the lognormal model (GBM) is a foundational tool in financial mathematics and underlies the Black–Scholes option pricing formula, it suffers from several well-documented drawbacks when confronted with real stock data. Our three stocks illustrate these limitations clearly.

Fat Tails

The most obvious failure is that empirical log-returns exhibit far heavier tails than the normal distribution assumed by GBM. In reality, large daily moves occur much more frequently than a standard bell curve predicts. An investor relying solely on this framework would systematically underestimate the likelihood of significant shocks to their portfolio.

Volatility is Not Constant

GBM assumes that volatility (σ) is constant over time. However, all three return series show clear evidence of volatility clustering: periods of calm alternate with periods of turbulence. This is a fundamental characteristic of actual markets that the model ignores. For K Wave Media in particular, volatility varies dramatically over the year, making a single σ estimate a poor summary of the stock's risk profile at any given moment.

Skewness

While the model assumes the returns are perfectly symmetric, all three stocks show clear non-zero skewness. K Wave Media and L'Oréal exhibit a negative skew, indicating that downside outliers tend to be more severe than upside gains, a phenomenon often called the "leverage effect". Because the GBM framework is inherently symmetric, it cannot account for this fundamental imbalance between potential risk and reward.

Drift and Volatility are Assumed Constant

The model uses a single estimated μ and σ fitted over the entire sample. In practice, the expected return and risk of a stock change over time due to macroeconomic conditions, company fundamentals, and market regimes. For K Wave Media, the enormous negative drift (−2.24 annualised) reflects a prolonged collapse; the GBM model simply propagates this forward indefinitely, generating unrealistic long-term forecasts. There is no mechanism within GBM for mean reversion or regime changes.

Continuous Prices and No Jumps

GBM generates continuous price paths. Real stocks, however, can experience sudden discontinuous jumps: earnings surprises, mergers and acquisitions, regulatory announcements, or market crashes can cause instantaneous price gaps that cannot be represented by a diffusion process alone.

Independence of Returns

GBM assumes that daily log-returns are independent and identically distributed. In practice, there is evidence of autocorrelation in return volatility (squared returns), and some degree of momentum in price levels, depending on the asset and time horizon. These dependencies are incompatible with the i.i.d. assumption.

Conclusion

We have analysed three stocks with very different characteristics: a highly volatile small-cap (K Wave Media), a mid-cap growth stock (Morgan) and a large-cap consumer staples company (L'Oréal). The lognormal model provides a tractable first approximation for stock price dynamics and correctly captures the exponential growth structure of prices and the multiplicative nature of returns. However, our empirical analysis confirms that it fails to reproduce the heavy tails, skewness, volatility clustering and potential for sudden jumps that characterise real financial data.

The lognormal model significantly underestimates risk for K Wave Media, as it cannot account for the extreme volatility clustering and negative skewness that drove its 93% price collapse. For Meta, the model's inherent symmetry fails to replicate the positive skewness observed in the data, missing the frequency of large upward moves. While L'Oréal is the most stable, it still exhibits a negative skew and left-tail bias that contradicts the model's assumption of perfectly symmetric returns.

The reliance on constant volatility and continuous price paths fails to represent the actual market "jumps" and shifting risk regimes seen across all three stocks.